

COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 1 : SLO 2 — De Morgan's Theorems (Practice)

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1. Match the Expressions (Remembering)

- 1. $(A + B)' \rightarrow$ c. $A'B'$
- 2. $(XY)' \rightarrow$ a. $X' + Y'$
- 3. $(A' + B')' \rightarrow$ b. AB (since $(A')(B') = AB$)
- 4. $(X'Y')' \rightarrow$ d. $X + Y$
- 5. $(A + B + C)' \rightarrow$ e. $A'B'C'$

2. Complete the Simplification (Understanding): $F = ((A + B)' \cdot C)'$

- Step 1: $((A + B)' \cdot C)'$ — given expression
- Step 2: $(A + B)'' + C'$ — De Morgan's theorem
- Step 3: $(A + B) + C'$ — double negation
- Step 4: $A + B + C'$ — final simplified expression

3. Simplify the Expressions (Applying)

- a. $F1 = (X' + Y' + Z')' = (X')(Y')(Z')' = X \cdot Y \cdot Z$
- b. $F2 = (A'B)' + C = (A)' + B' + C = A + B' + C$
- c. $F3 = ((A + B)' + C')' = ((A + B))' \cdot (C')' = (A + B) \cdot C = AC + BC$

4. Find the Error (Analyzing): $F = (A(B + C))'$

- The error occurred in Step 2 (the student wrote $A'(B + C)'$).
- Mistake: De Morgan's theorem states $(XY)' = X' + Y'$. Treating A and $(B + C)$ as the two terms, the complement of the product must become a SUM of the complements, not a product. The student kept multiplication after negating.
- Correct simplification: $F = (A(B + C))' = A' + (B + C)' = A' + B'C'$

5. Create a Problem (Creating)

- a. Created expression: $F = (X' + YZ)'$
- b. $F = (X')' \cdot (YZ)'$ (De Morgan's theorem)
- $F = X \cdot (Y' + Z')$ (double negation + De Morgan's theorem)
- $F = XY' + XZ'$ (distributive law)